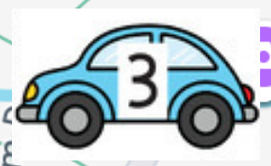
Parking Functions



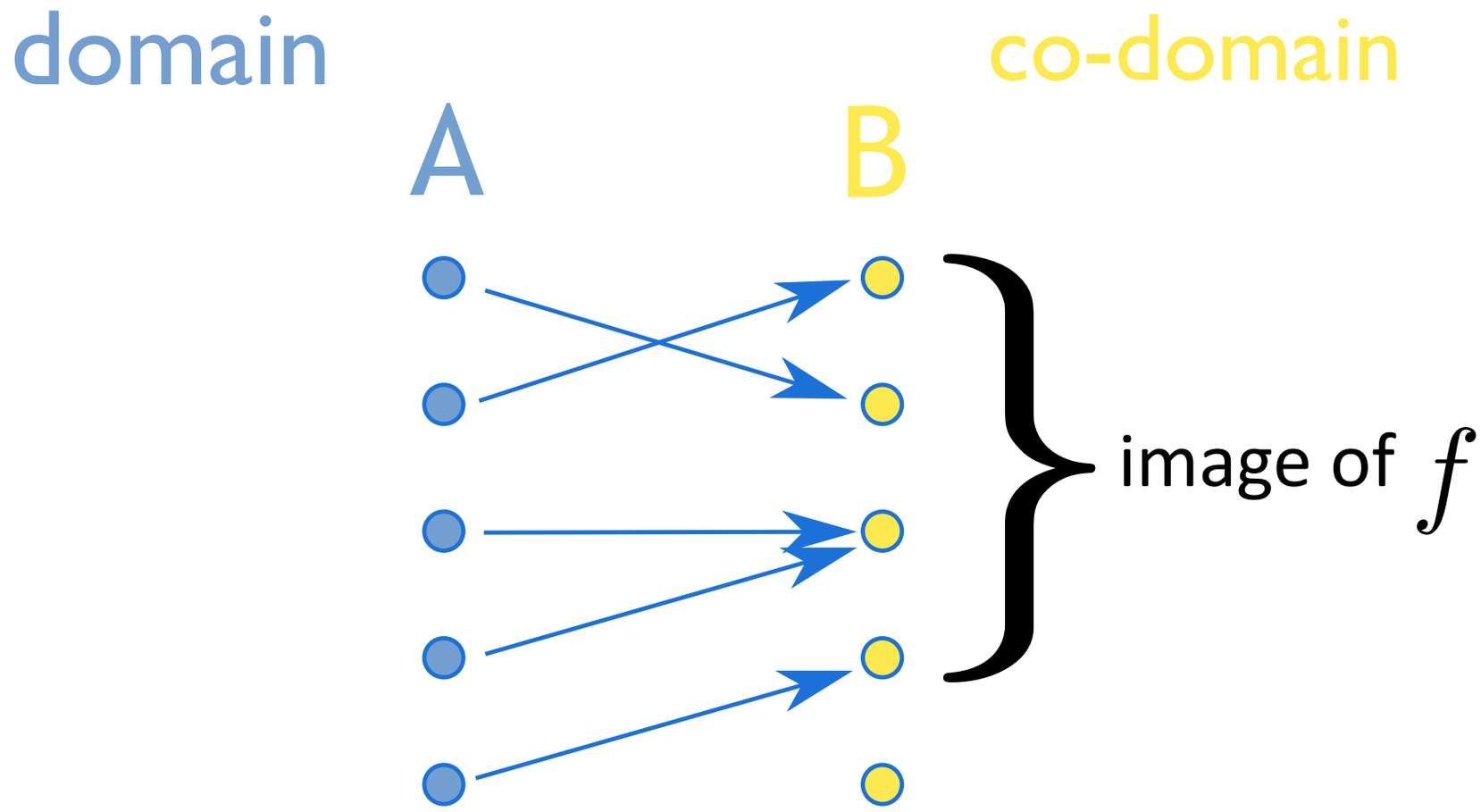




Biomath Seminar
Greg Conradi Smith
2025 September 12

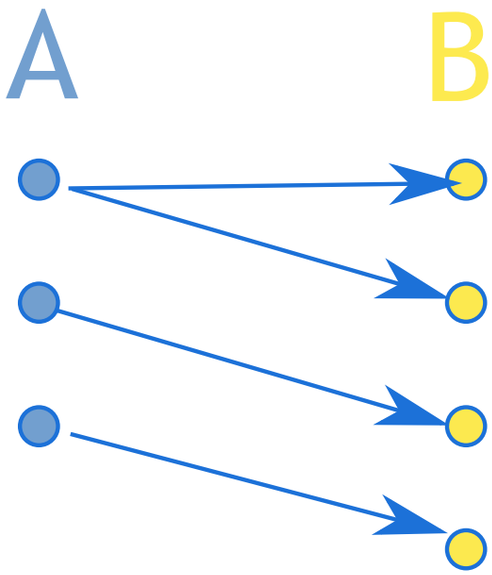
What is a **bijection**?

$$f : A \longrightarrow B$$



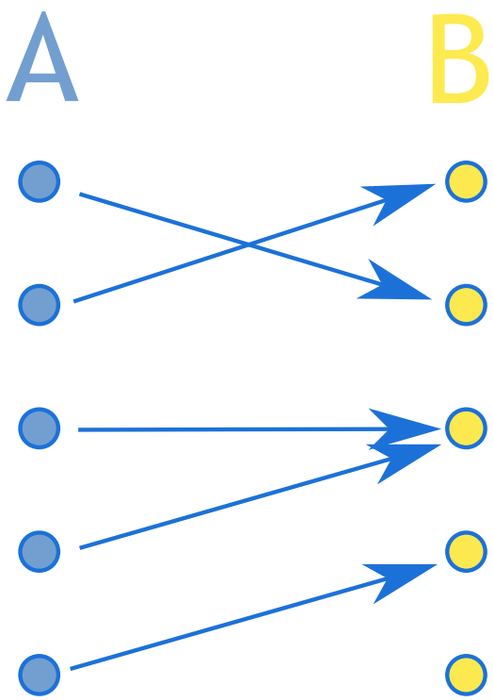
General Function

$$f : A \longrightarrow B$$

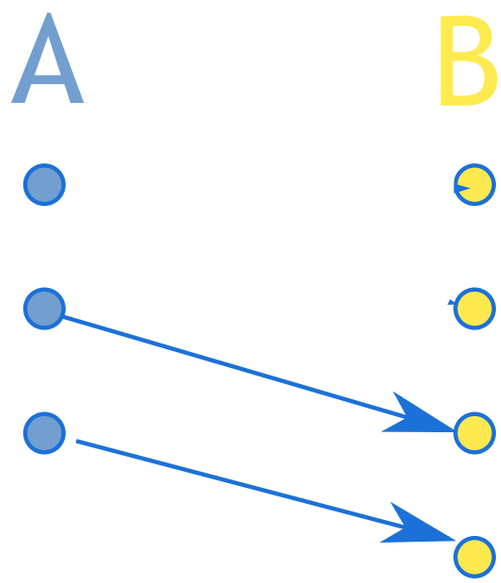


NOT a Function

A has many B



Function



NOT a Function

every element of A must be mapped

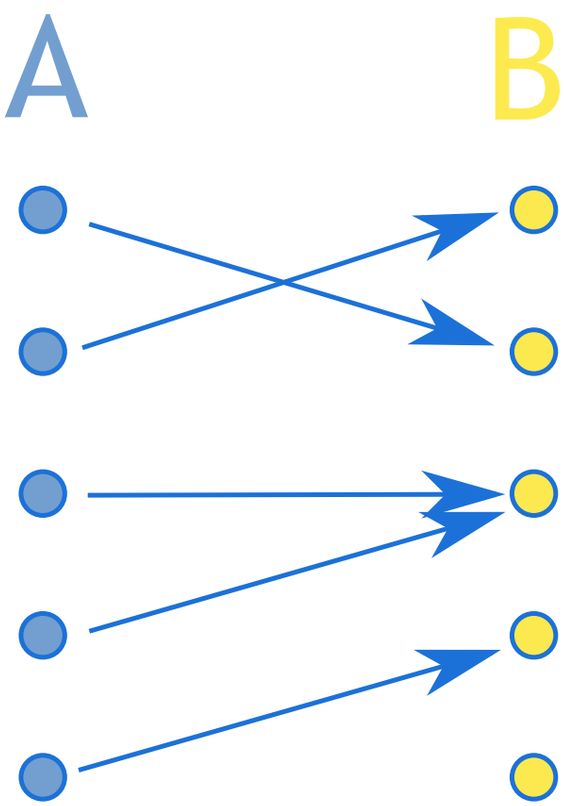
How many distinct functions from A to B?

domain

#A = n

co-domain

#B = m



?

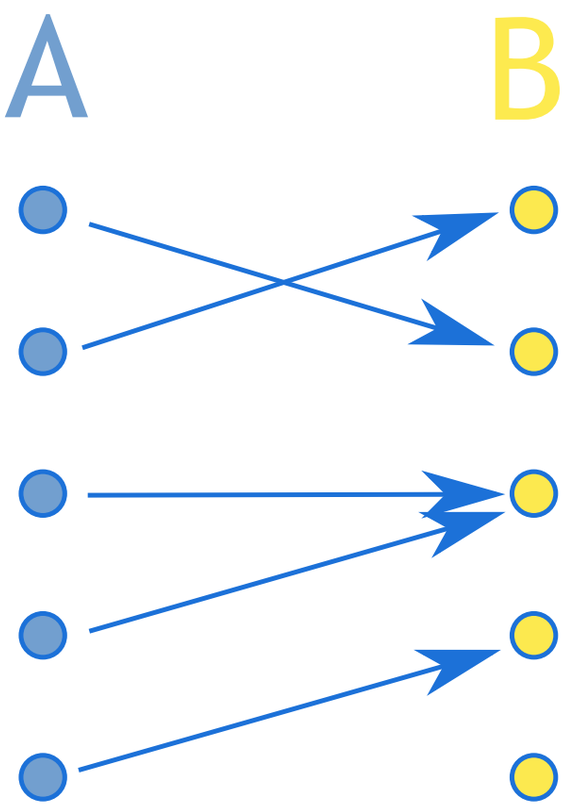
How many distinct functions from A to B?

domain

#A = n

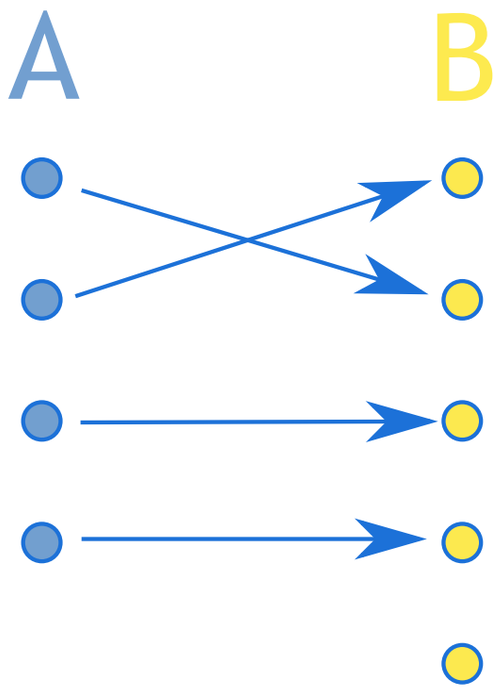
co-domain

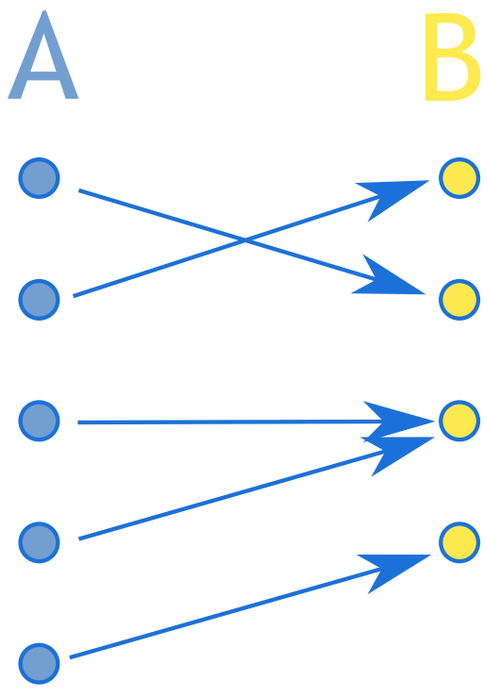
#B = m

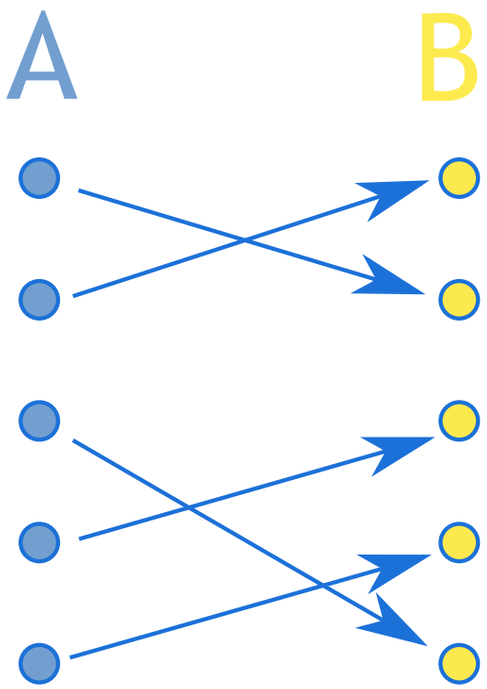


$$m^n$$

$$f : A \longrightarrow B$$







Injective
(not surjective)

B can't have many A

Surjective
(not injective)

Every B has some A

Bijective
(injective, surjective)

A to B, perfectly

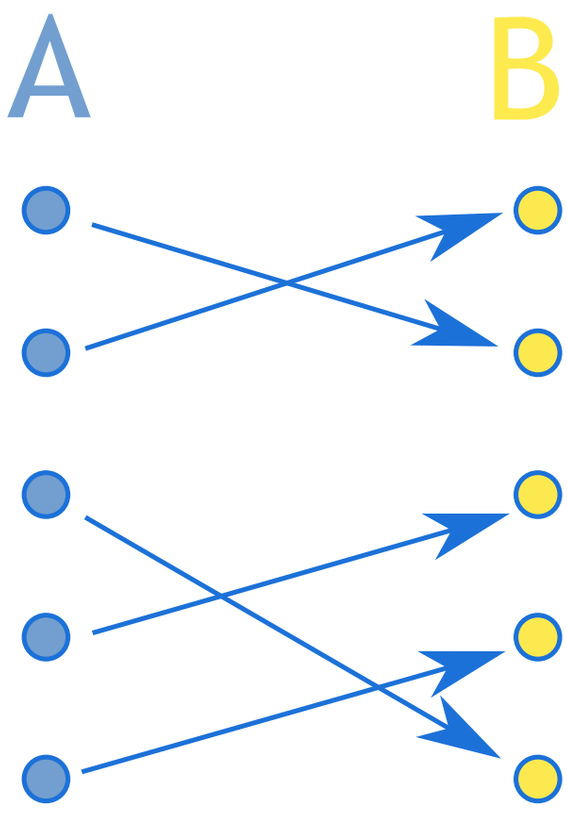
"one-to-one"

"onto"

https://www.mathsisfun.com/sets/injective-surjective-bijective.html

What is a **bijection**?

A **bijection** is a **map** between two sets that is **one-to-one** and **onto**

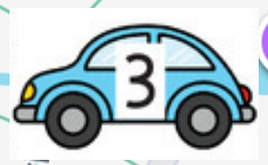


Bijective

A bijective map is **invertible**

Two sets with a bijective map must have the same **cardinality**

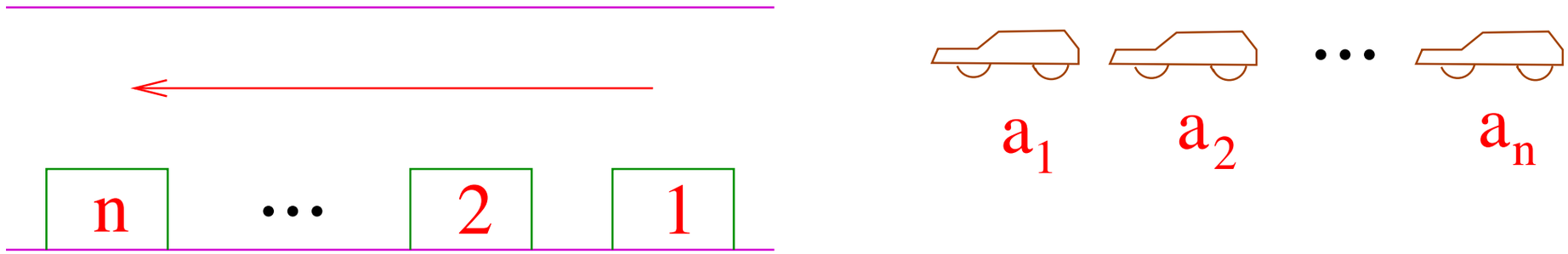
What is a parking function?





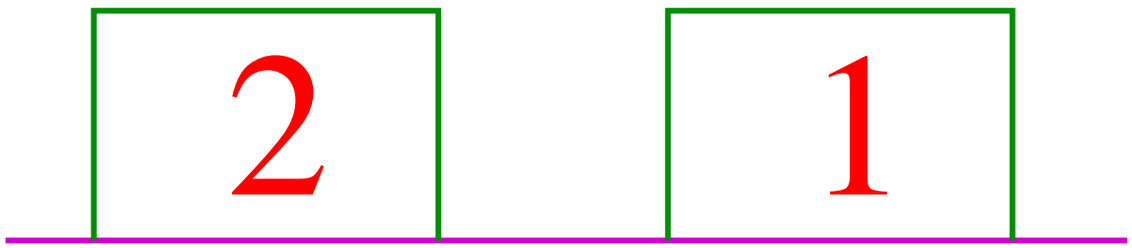


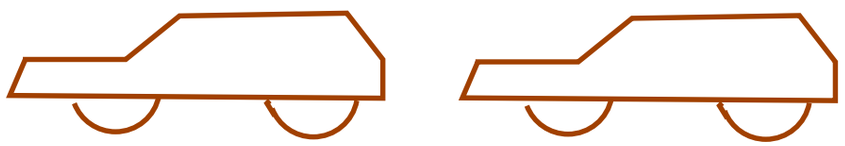
What is a **parking function**?



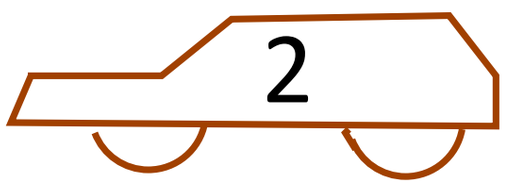
Car C_i prefers space a_i. If a_i is occupied, then C_i takes the next available space. We call $(a_1, \ldots, a_n)$ a **parking function** (of length n) if all cars can park.

What are the parking functions of length 2?

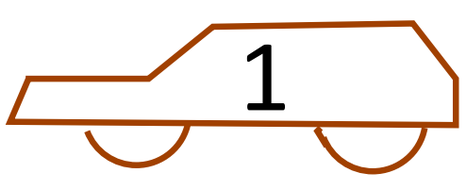


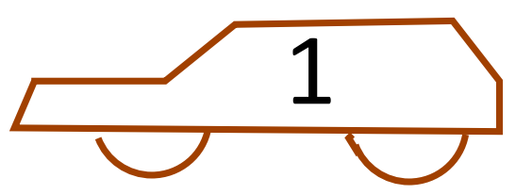


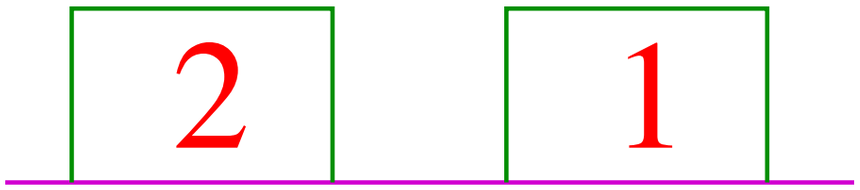
a_1 a_2



a_1









a_1 a_2

parking functions of length 2



a_2



a_1

1 1





a_2



a_1

1 2





a_1



a_2

2 1





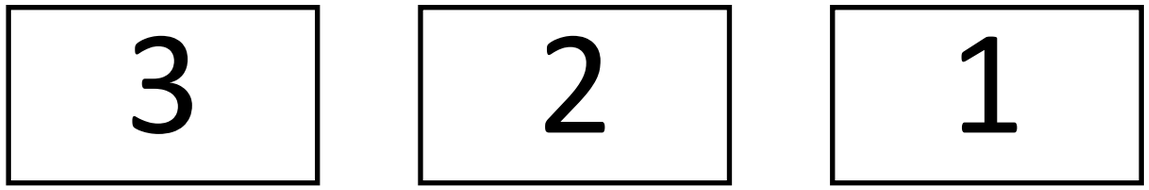
a_1

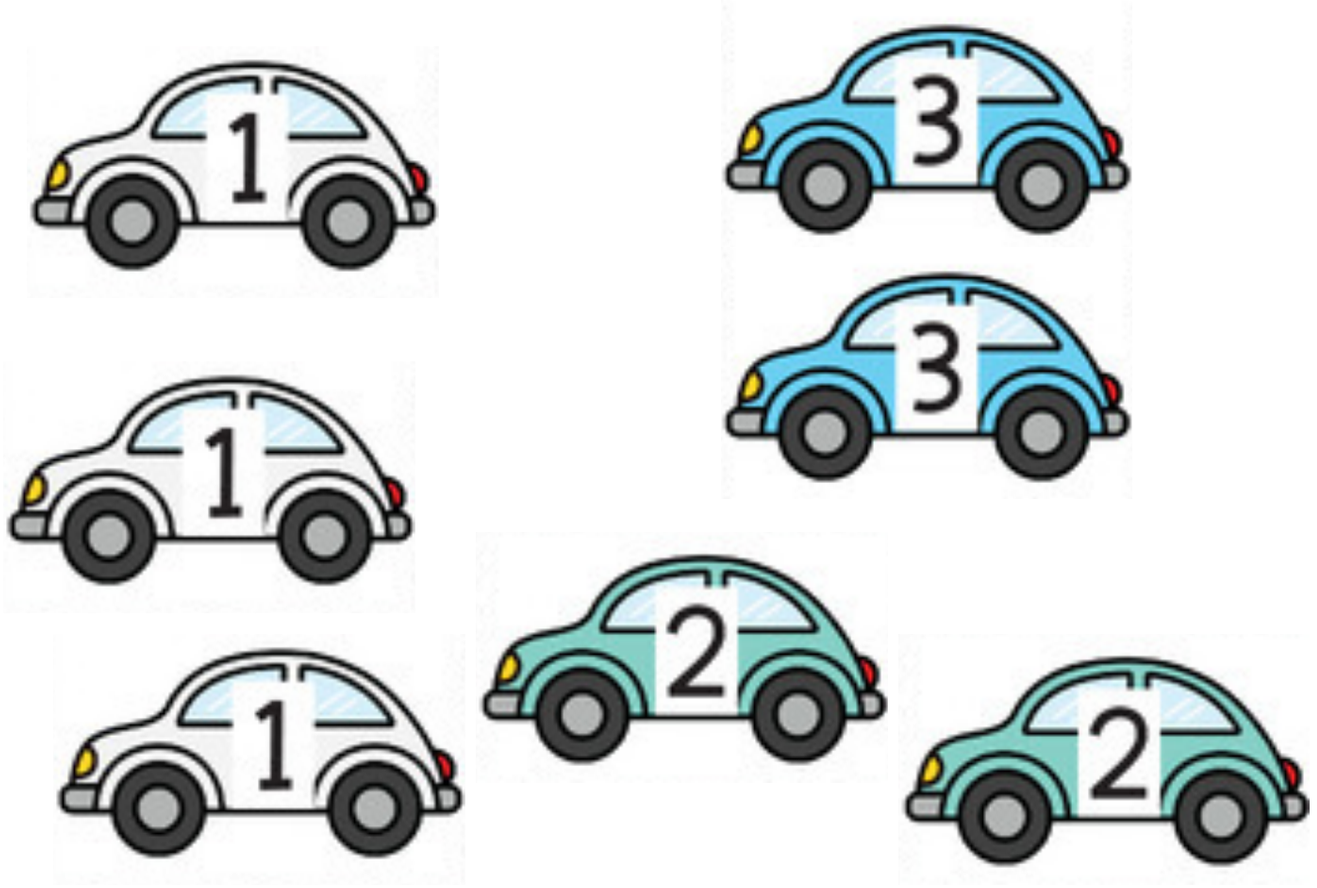


a_2

~~2 2~~







parking functions of length 3

111	**211**	**311**
112	**212**	**312**
113	**213**	313
121	**221**	**321**
122	222	322
123	223	323
131	**231**	331
132	232	332
133	233	333

What are these sequences callled parking “functions”?

a=(1,3,1)



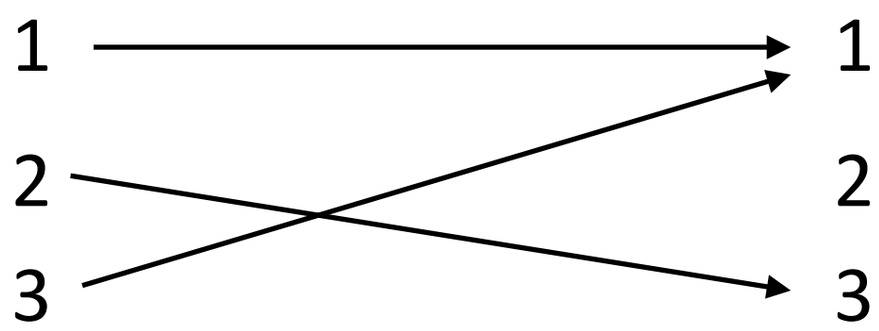
first car second car third car

position in car line

{1,2,3}

parking preference

{1,2,3}



How to determine if a **sequence** is a **parking function**?

Easy: Let $\alpha = (a_1, \ldots, a_n) \in \mathbb{P}^n$. Let $b_1 \leq b_2 \leq \cdots \leq b_n$ be the increasing rearrangement of α. Then α is a parking function if and only $b_i \leq i$.

a=(33151) **b**=(11335) b1=1≤1 b2=1≤2 b3=3≤3 b4=3≤4 b5=5≤5

a=(35151) **b**=(11355) b1=1≤1 b2=1≤2 b3=3≤3 b4=5≤4 b5=5≤5

Corollary. *Every permutation of the entries of a parking function is also a parking function.*

111 211 311

112 212 312

113 213 313

121 221 321

122 222 322

123 223 323

131 231 331

132 232 332

133 233 333

111

112 211 121

113 131 311

122 212 221

123 132 213

231 321 312

How many equivalence classes of parking functions of length 4?

Start with all the weakly increasing sequences…

1111	1133	1234	2223	2344
1112	1134	1244	2224	2444
1113	1144	1333	2233	3333
1114	1222	1334	2234	3334
1122	1223	1344	2244	3344
1123	1224	1444	2333	3444
1124	1233	2222	2334	4444

Did I get them all?

What is the general formula for the number of weakly increasing sequences of 1..n of length n

$$\boxed{\# = \binom{2n-1}{n} = \binom{2n-1}{n-1}}$$

Reason: A weakly increasing sequence $(x_1 \le \cdots \le x_n)$ from $\{1, \ldots, n\}$ is the same as choosing counts $c_1, \ldots, c_n \ge 0$ (how many 1s, 2s, ..., ns) with

$$c_1 + \cdots + c_n = n.$$

By stars and bars, the number of nonnegative integer solutions is

$$\binom{n + (n-1)}{n} = \binom{2n-1}{n}.$$

$$\binom{2 \cdot 4 - 1}{4} = \binom{7}{4} = \frac{7!}{4!3!} = \frac{7 \cdot 6 \cdot 5}{3!} = 35$$ ✅

Now eliminate those for which b1>1

1111	1133	1234	2223	2344
1112	1134	1244	2224	2444
1113	1144	1333	2233	3333
1114	1222	1334	2234	3334
1122	1223	1344	2244	3344
1123	1224	1444	2333	3444
1124	1233	2222	2334	4444

And also eliminate those for which b2>2

1111	1133	1234
1112	1134	1244
1113	1144	1333
1114	1222	1334
1122	1223	1344
1123	1224	1444
1124	1233	

And also eliminate those for which b3>3

1111	1133	1234
1112	1134	1244
1113	1144	
1114	1222	
1122	1223	
1123	1224	
1124	1233	

For none of the remaining sequences is b4>4

1111 1133 1234

1112 1134

1113

1114 1222

1122 1223

1123 1224

1124 1233

Now consider all permutations and count.

We should get:

$$(n+1)^{n-1} = 5^3 = 125$$

parking functions.

1+1+1+1 | 1234 | 4!=24

1+1+2 | 1123 | 1124 | 1134 | 1223 | 1224 | 1233 | 4!/(1!1!2!)=12

6 x 12 = 72

2+2 | 1122 | 1133 | 4!/(2!2!)=6 | 2 x 6 = 12

3+1 | 1112 | 1113 | 1114 | 1222 | 4!/(1!3!)=4 | 4 x 4 = 16

4 | 1111 | 4!/4!=1 | 1 x 1 = 1

24 + 72 + 12 + 16 + 1 = 125

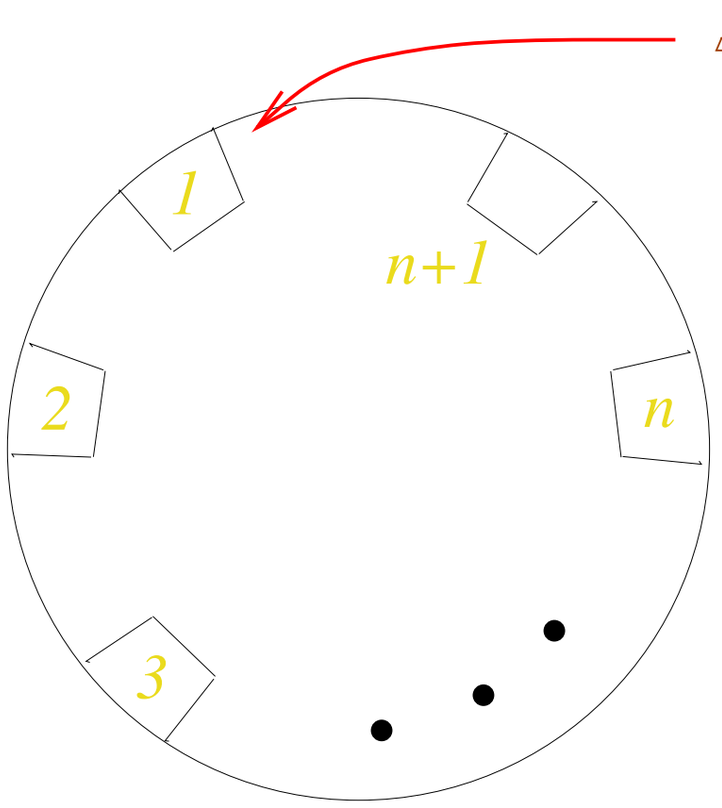
$(4+1)^{(4-1)} = 5^3 = 125$

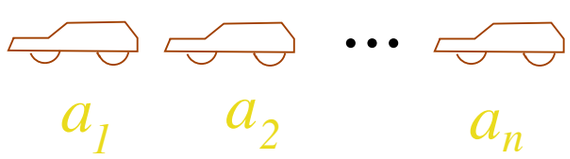


Enumeration of parking functions

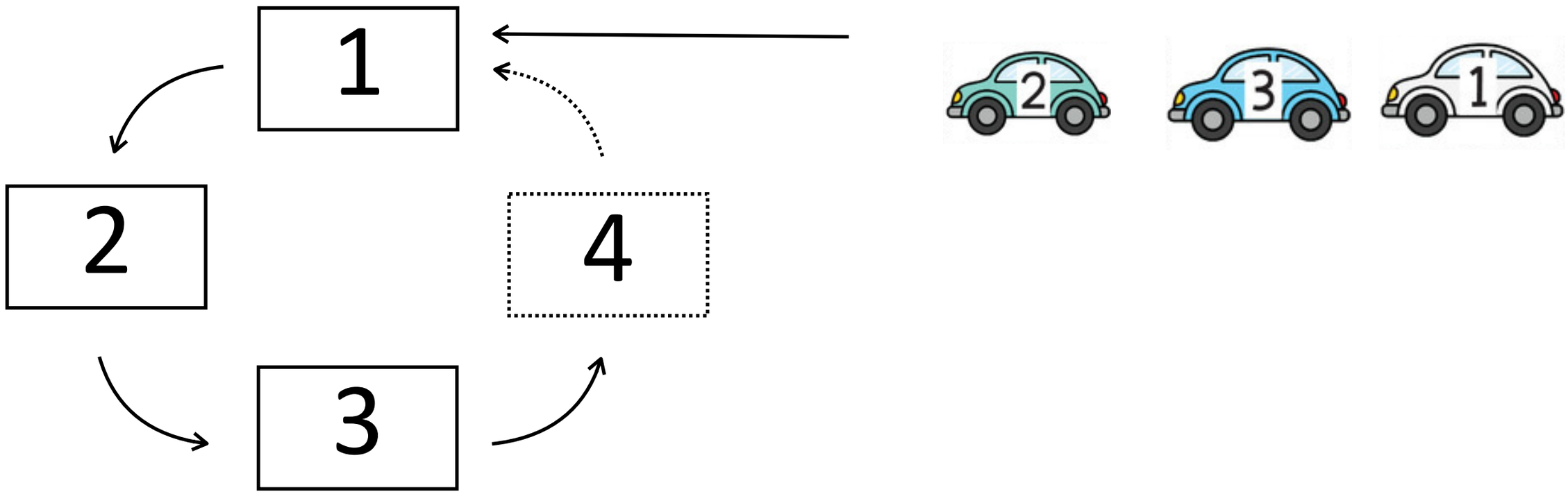
Theorem (**Pyke**, 1959; **Konheim and Weiss**, 1966). *Let $f(n)$ be the number of parking functions of length n. Then* $\boldsymbol{f(n) = (n+1)^{n-1}}$.

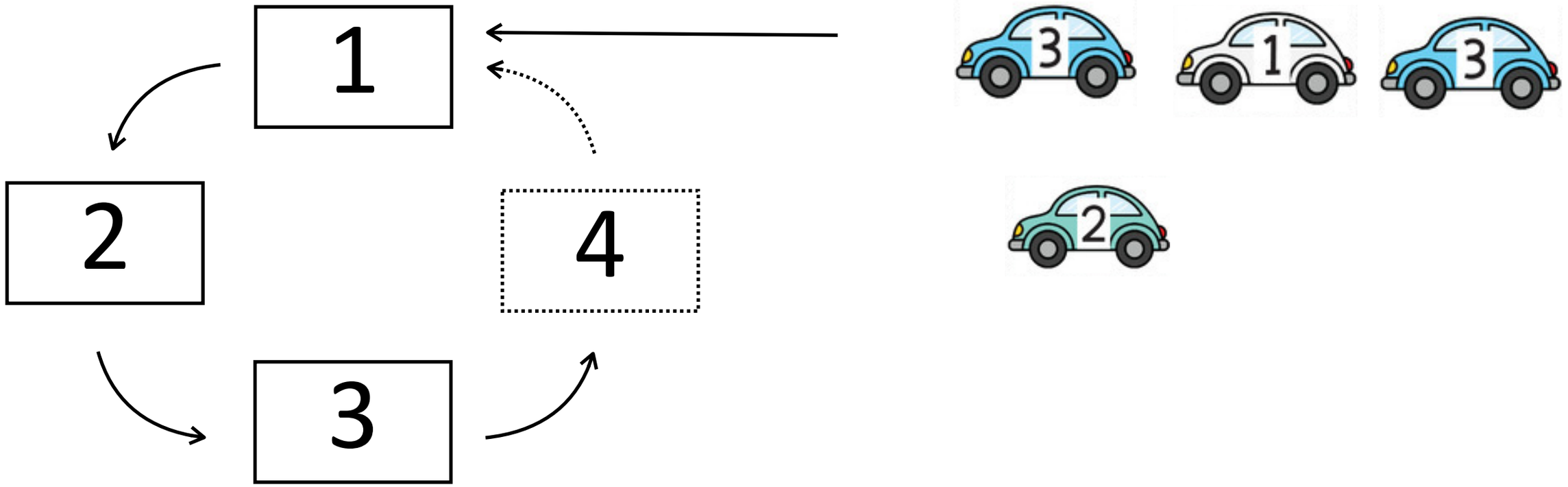
Proof (**Pollak**, c. 1974). Add an additional space $n+1$, and arrange the spaces in a circle. Allow $n+1$ also as a preferred space.

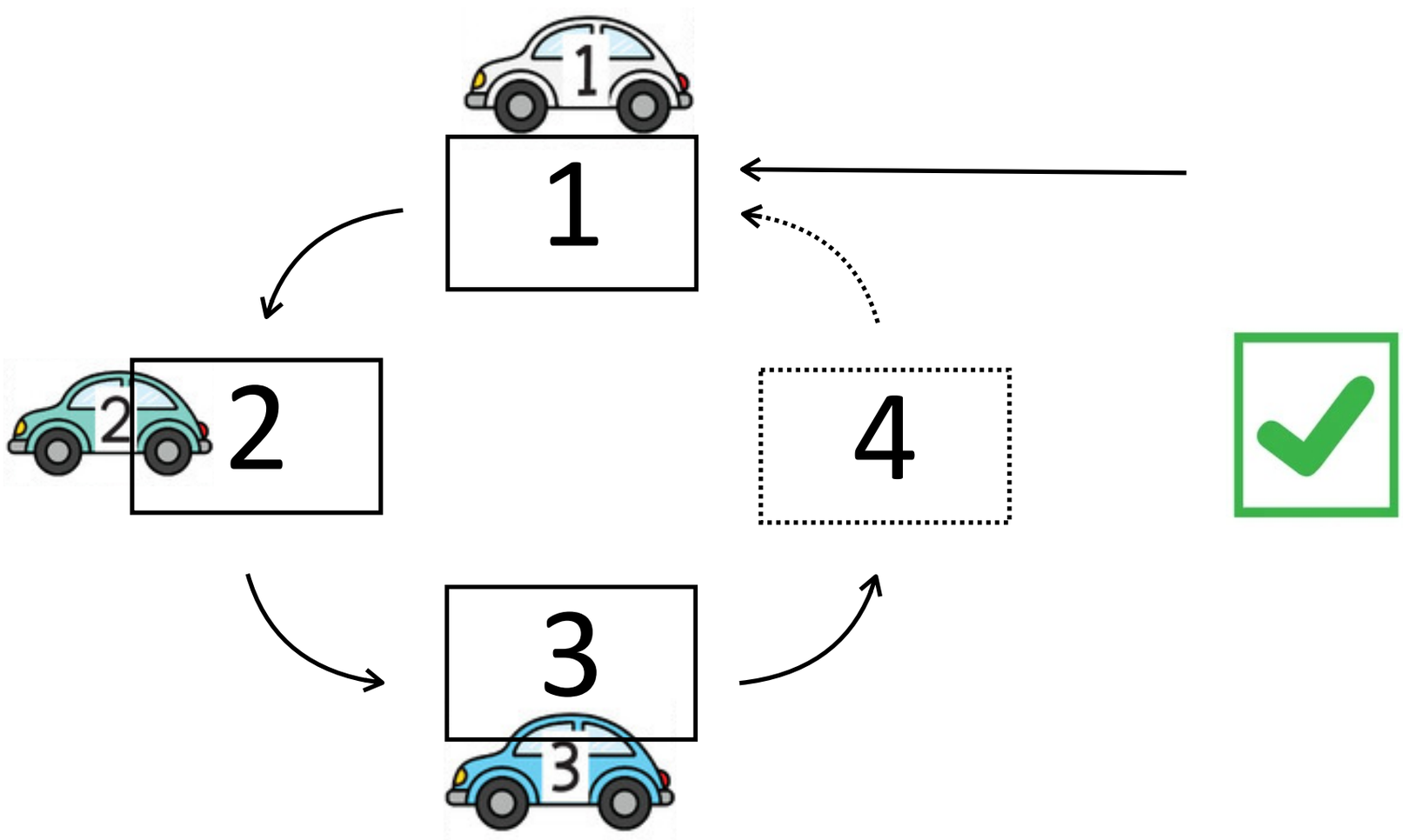


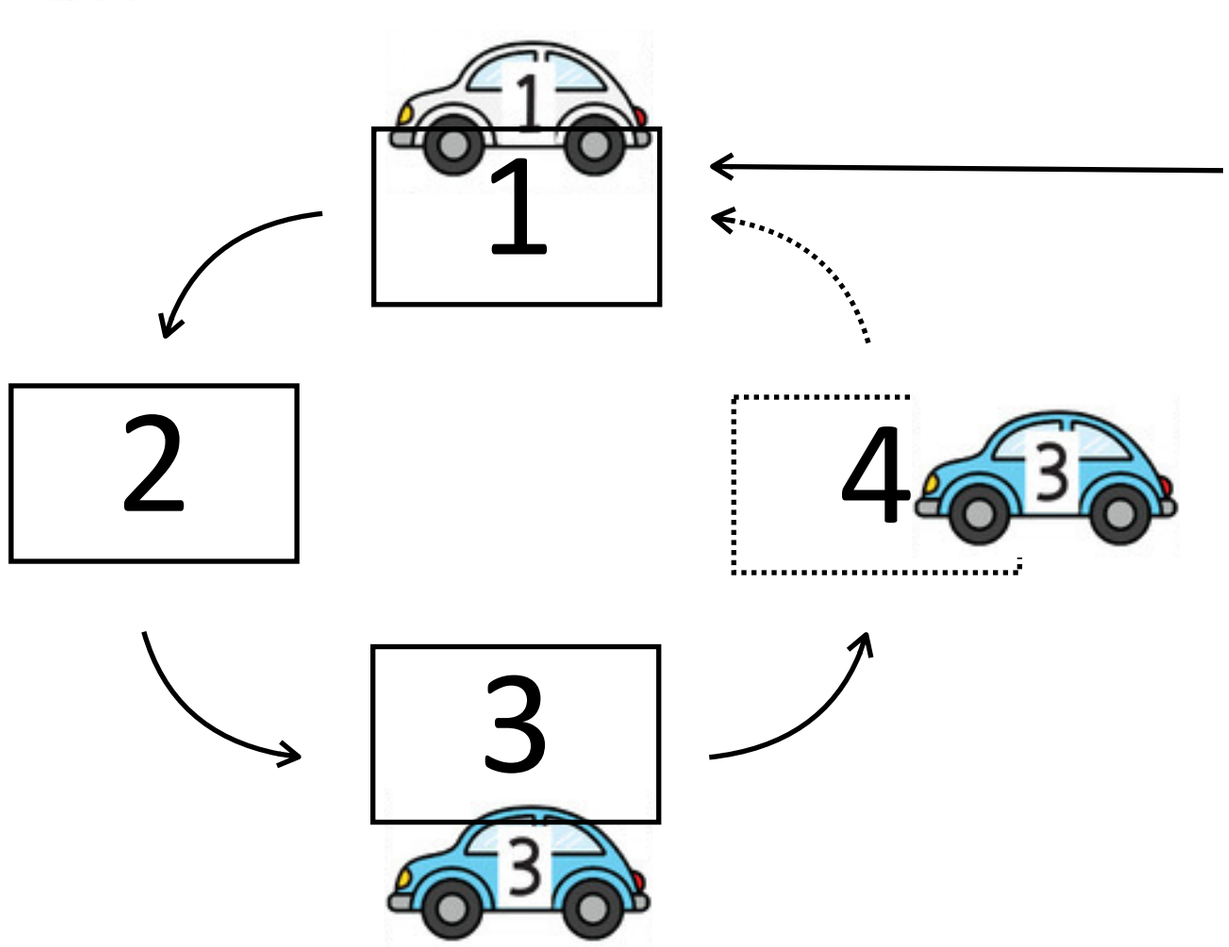


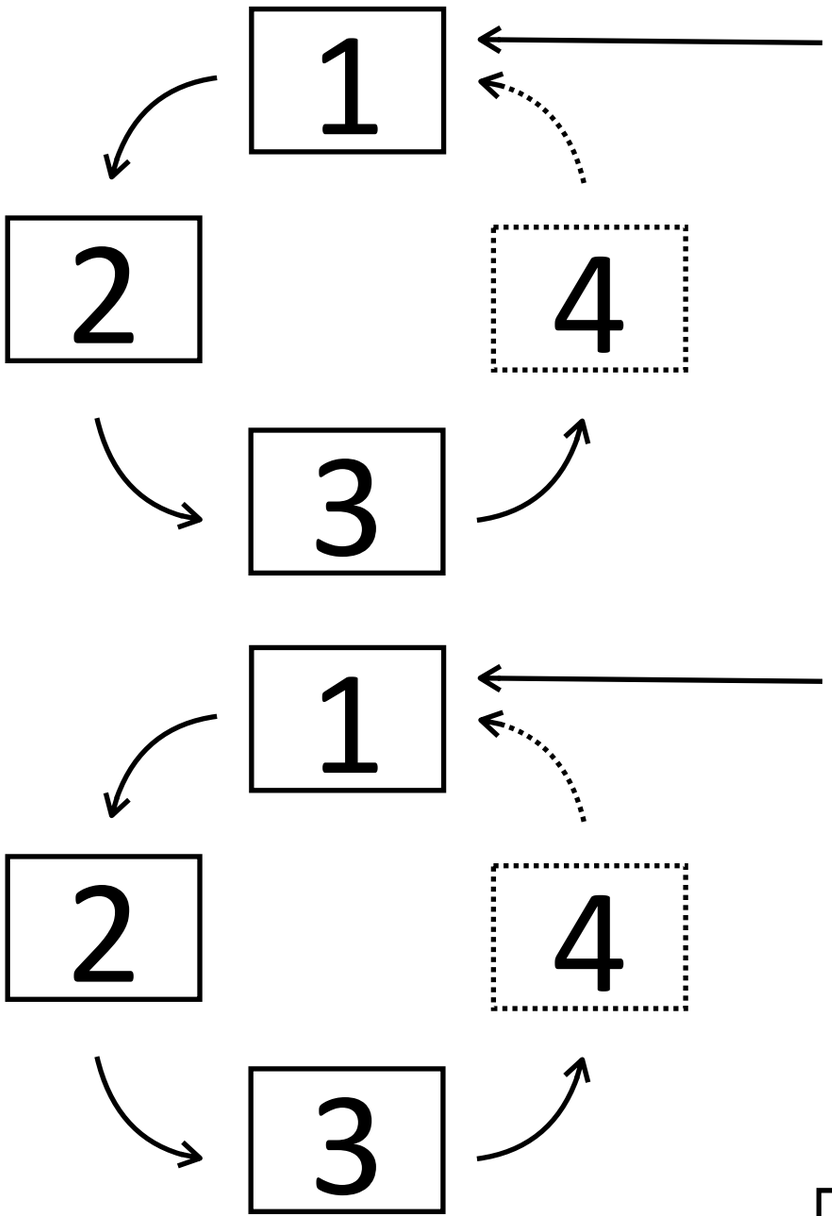
Now every car will park.
(a1,a2,…,an) is parking function if space n+1 remains empty


1
2
3
4
2
3
1

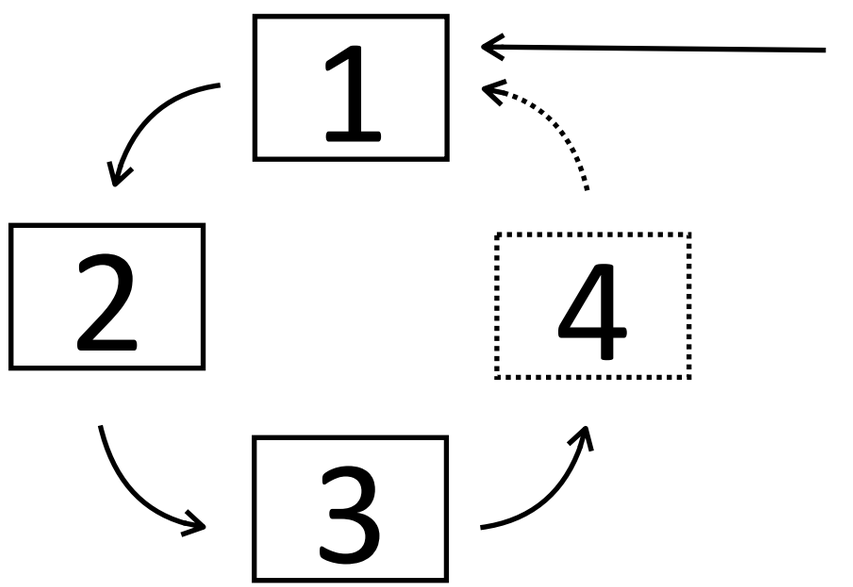

1
2
3
4
3
1
3
2

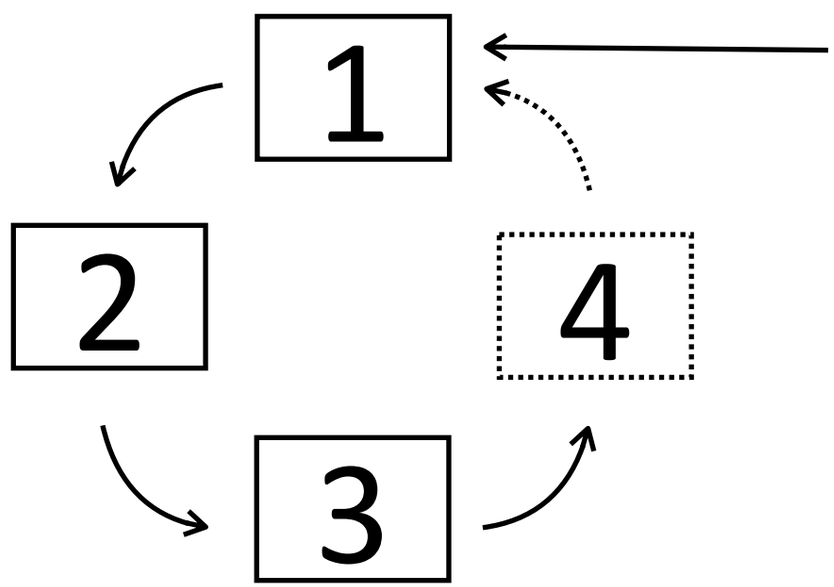

1
1
2
2
4
3
3

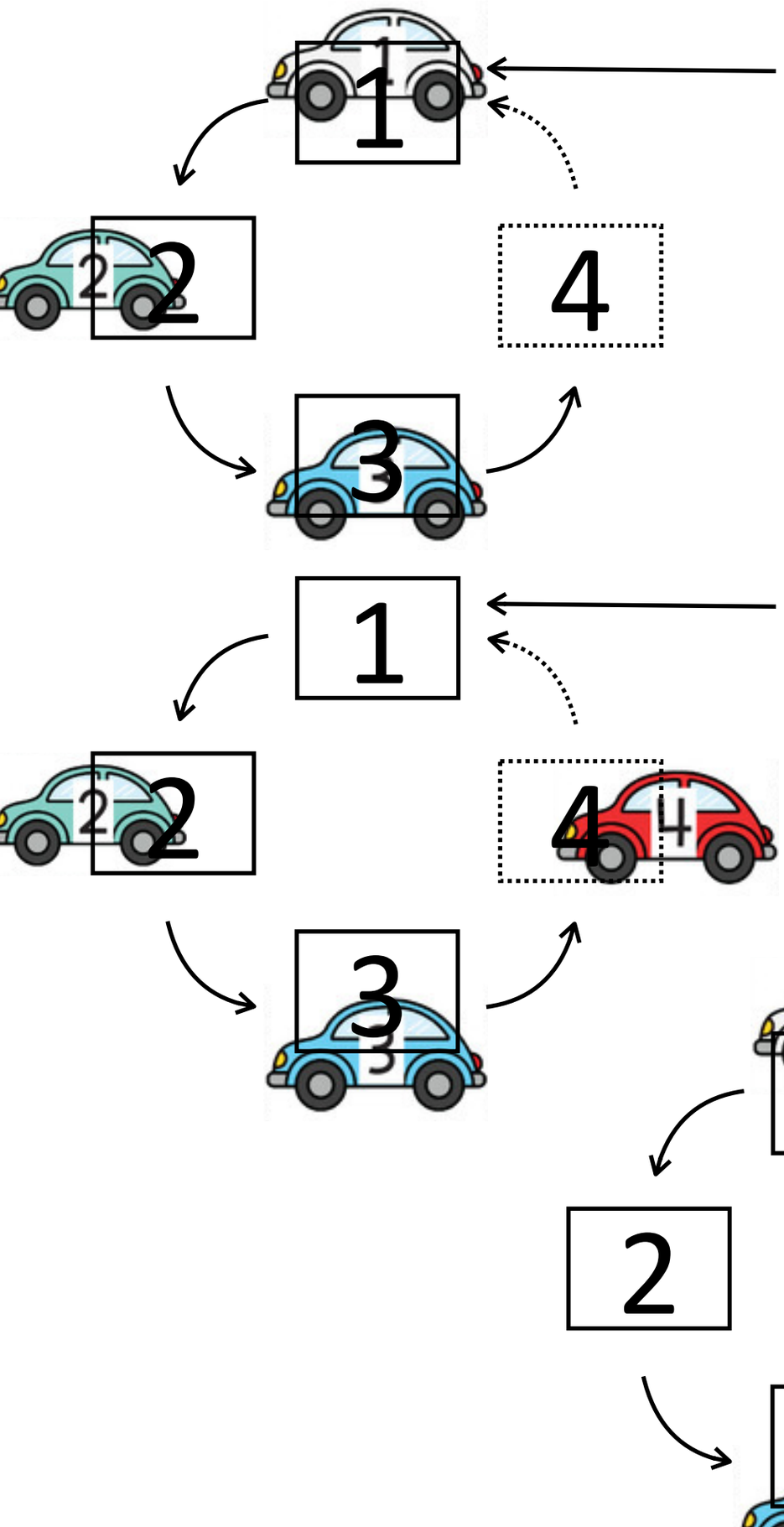

1
1
2
4
3
3
3


1
2
4
3


2
3
1
3
4
2
4
1
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2
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1
2
4
3


1
2
4
3

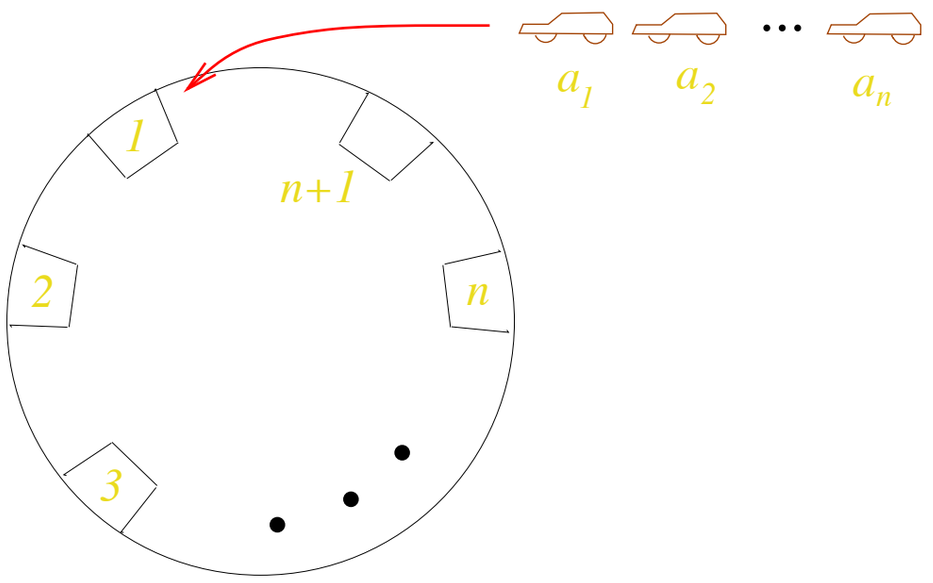


In this modified problem, adding j to each ai gives a final result shifted counterclockwise by j parking spots

Enumeration of parking functions

Theorem (**Pyke**, 1959; **Konheim and Weiss**, 1966). *Let $f(n)$ be the number of parking functions of length n. Then $f(n) = (n+1)^{n-1}$.*

Proof (**Pollak**, c. 1974). Add an additional space $n+1$, and arrange the spaces in a circle. Allow $n+1$ also as a preferred space.



Now all cars can park, and there will be one empty space. α is a parking function $\Leftrightarrow$ if the empty space is $n+1$. If $\alpha = (a_1, \ldots, a_n)$ leads to car C_i parking at space p_i, then $(a_1 + j, \ldots, a_n + j)$ (modulo $n+1$) will lead to car C_i parking at space $p_i + j$. Hence exactly one of the vectors

$$(a_1 + i, a_2 + i, \ldots, a_n + i) \text{ (modulo } n+1)$$

is a parking function, so

$$f(n) = \frac{(n+1)^n}{n+1} = (n+1)^{n-1}.$$

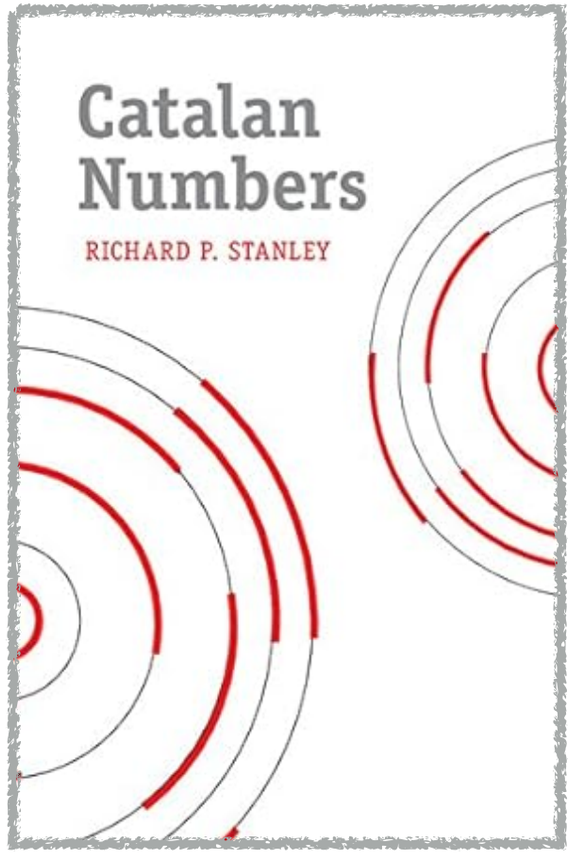
$$\#\left\{\begin{array}{c}\text{weaking increasing}\\ \text{parking functions}\\ \text{of length } n\end{array}\right\} = C_n = \frac{1}{n+1}\binom{2n}{n}$$

The first Catalan numbers for $n = 0, 1, 2, 3, \ldots$ are

1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796, 58786, ... (sequence A000108 in the OEIS).

OEIS:A000108

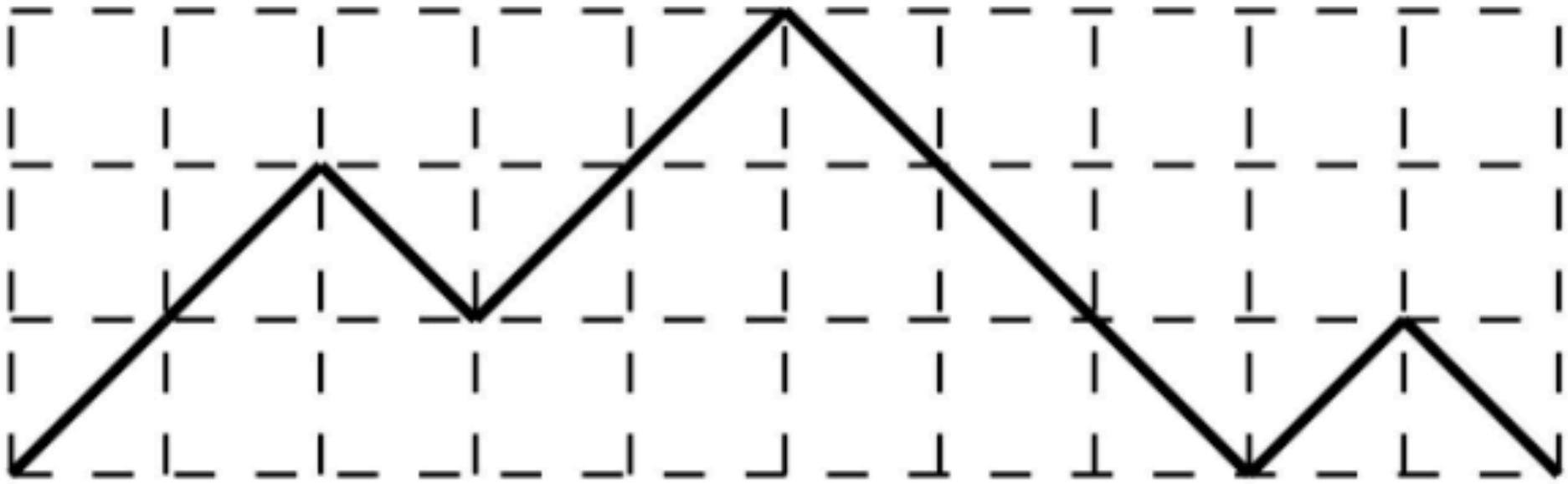
https://oeis.org/A000108



How does one prove something like this?

Dyck path of length 2n has n up steps and n down steps

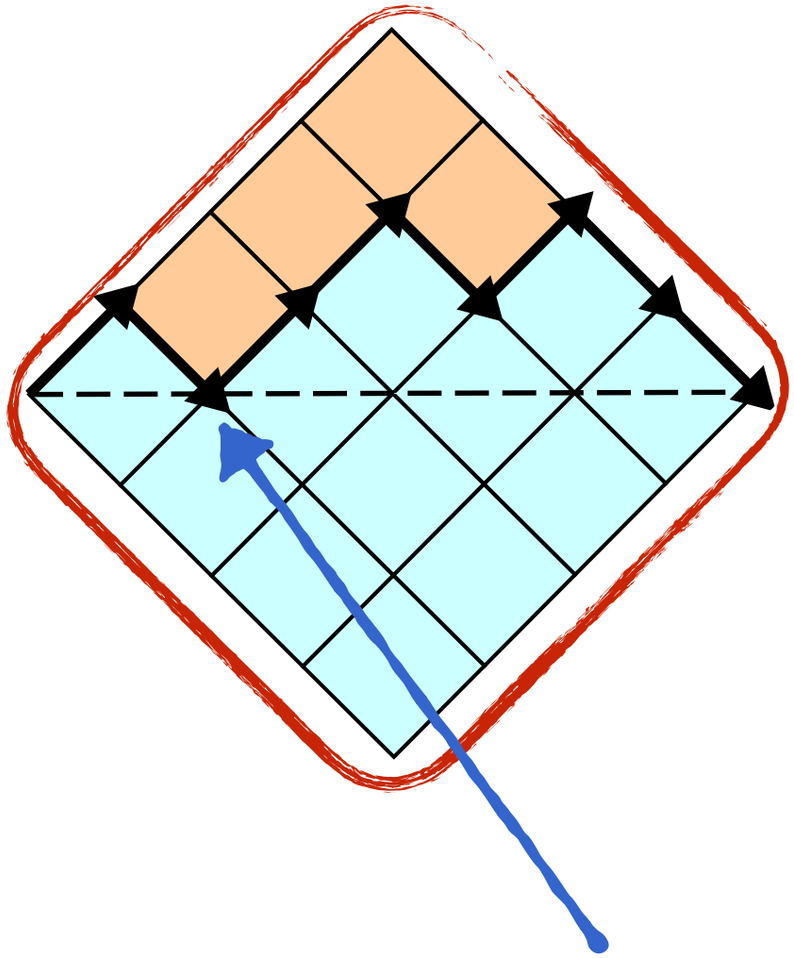
This is a Dyck path of length 10 (n=5)



There are C_n Dyck paths of length 2n

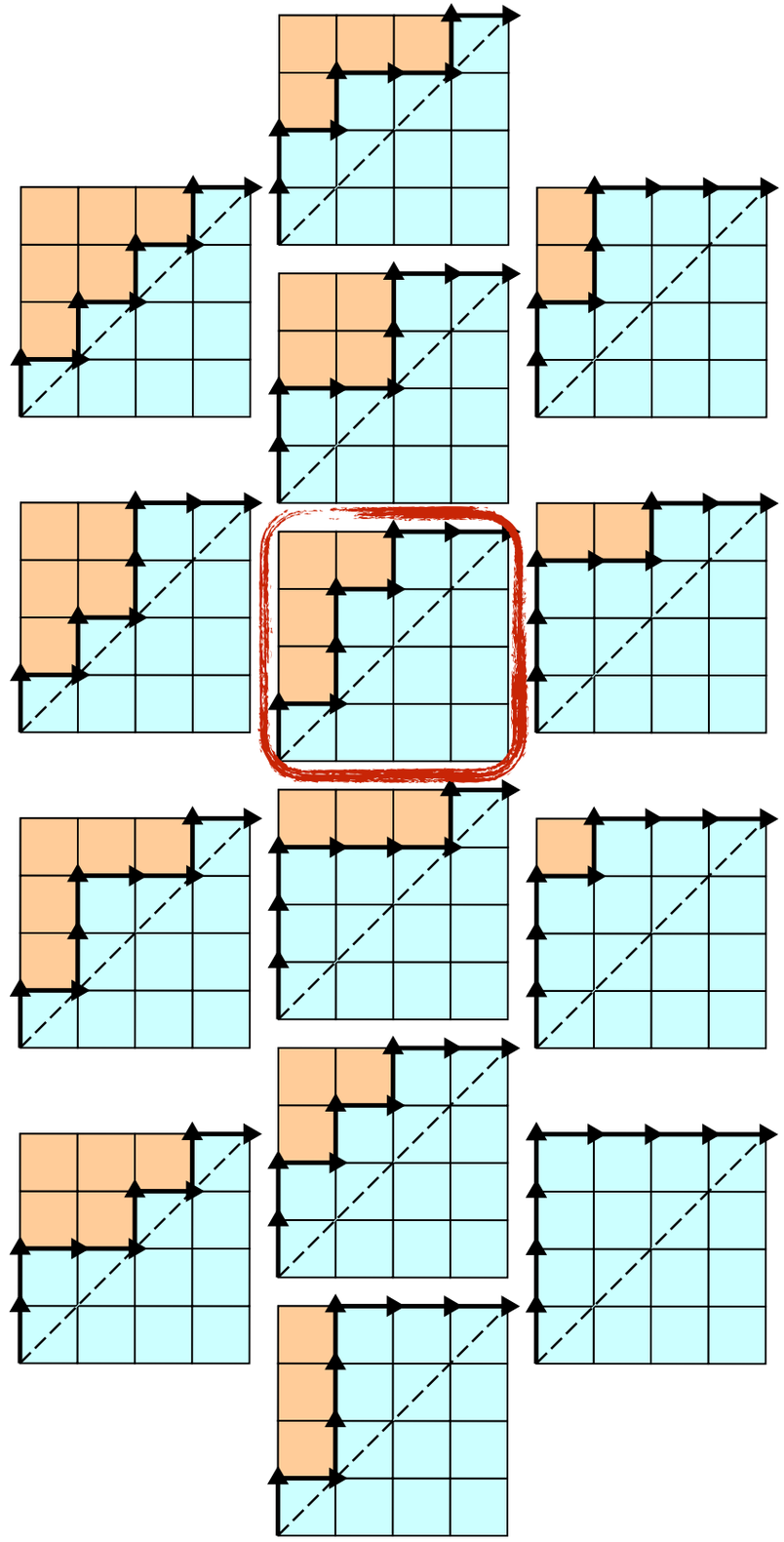
For example, when n = 4 …

Dyck path of length 8



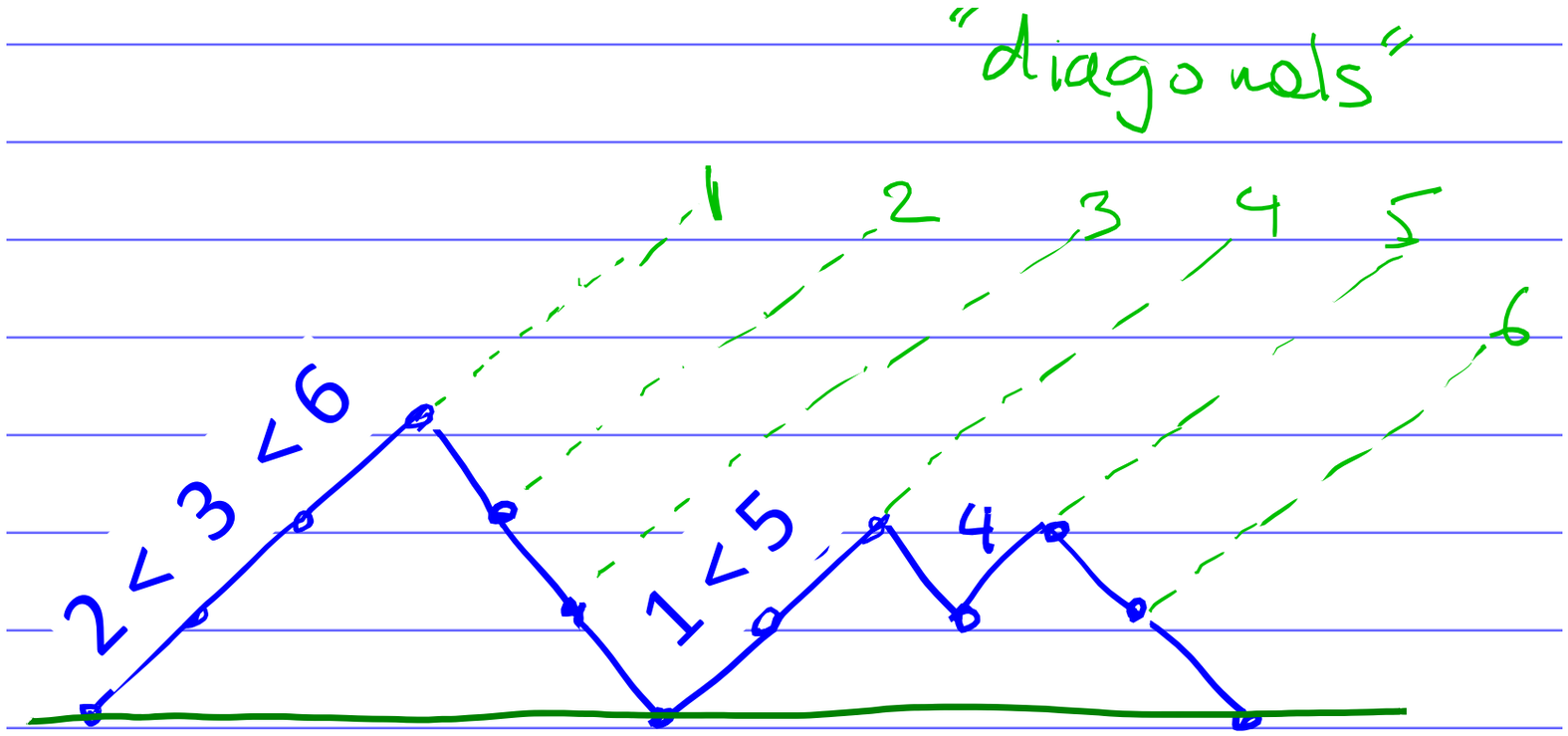
The Catalan numbers satisfy the recurrence relations

$$C_0 = 1 \quad \text{and} \quad C_n = \sum_{i=1}^{n} C_{i-1} C_{n-i} \quad \text{for } n > 0$$



$C_4 = 14$

labelled Dyck paths $\longleftrightarrow$ parking functions (1:1)



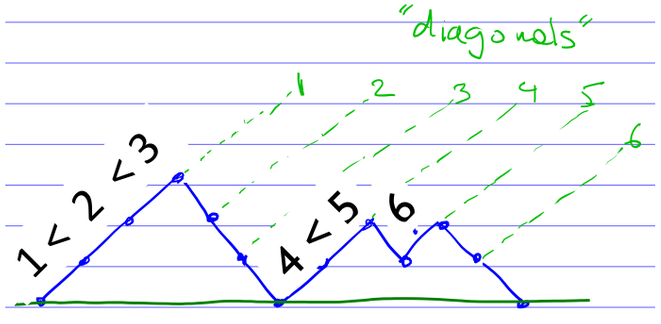
i=(1,2,3,4,5,6)

a=(4,1,1,5,4,1)

b=(1,1,1,4,4,5)

≤

i=(1,2,3,4,5,6)



A clean, standard map sends a parking function $a = (a_1, \ldots, a_n)$ to a (labeled) Dyck path of semilength n by using the value-multiplicities as the vertical run lengths.

Map (parking function $\longrightarrow$ Dyck path)

1. Let b be the nondecreasing rearrangement of a.
2. For each $k \in \{1, \ldots, n\}$, let $m_k = \#\{i : a_i = k\} = \#\{j : b_j = k\}$.
3. Form the Dyck word

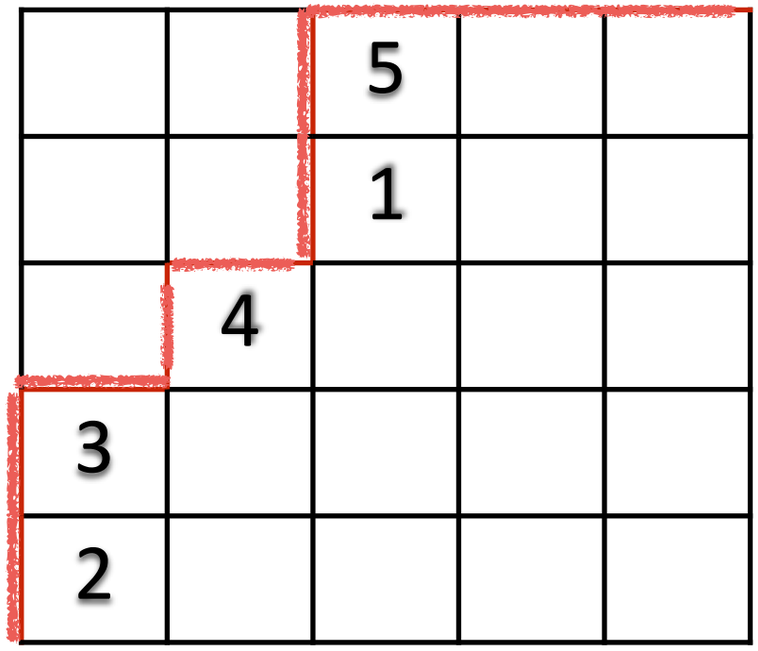
$$\underbrace{N \cdots N}_{m_1} E \; \underbrace{N \cdots N}_{m_2} E \; \cdots \; \underbrace{N \cdots N}_{m_n} E,$$

i.e., columns of m_k north steps, separated by one east step.
Because a is a parking function, $\sum_{i=1}^{k} m_i \geq k$ for all k, so the path never drops below the diagonal.

(Optional labels, giving the classical *labeled* Dyck path representation.)
For each k, take the set $S_k = \{i : \ a_i = k\}$ ordered increasingly, and place those labels on the m_k north steps in the k-th column from bottom to top. Labels then increase within each column.

Tiny example

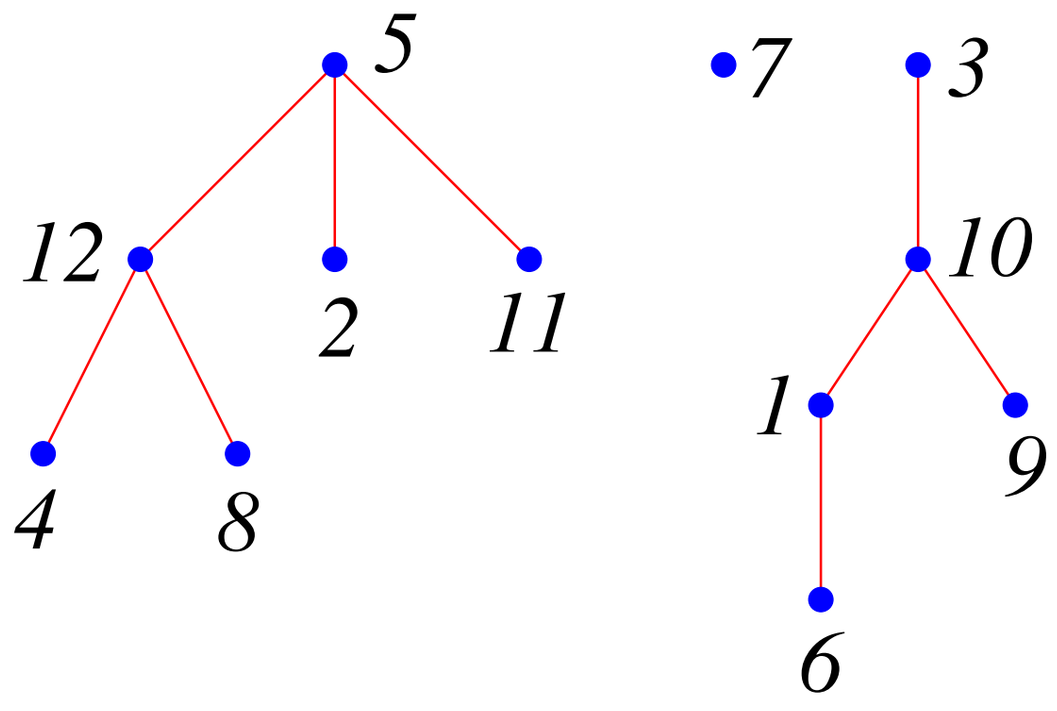
$a = (3, 1, 1, 2, 3) \Rightarrow b = (1, 1, 2, 3, 3)$.
Multiplicities: $(m_1, m_2, m_3, m_4, m_5) = (2, 1, 2, 0, 0)$.
Dyck word: $NN\,E\,N\,E\,NN\,E\,E\,E$.
Labels by columns: column 1 holds labels $\{2, 3\}$ (bottom–top),
column 2 holds $\{4\}$, column 3 holds $\{1, 5\}$.



Labeled Dyck paths of length 2n
are in bijection with
parking functions of length n.

$$\#\left\{\begin{array}{c}\text{weaking increasing}\\ \text{parking functions}\\ \text{of length } n\end{array}\right\} = C_n = \frac{1}{n+1}\binom{2n}{n}$$

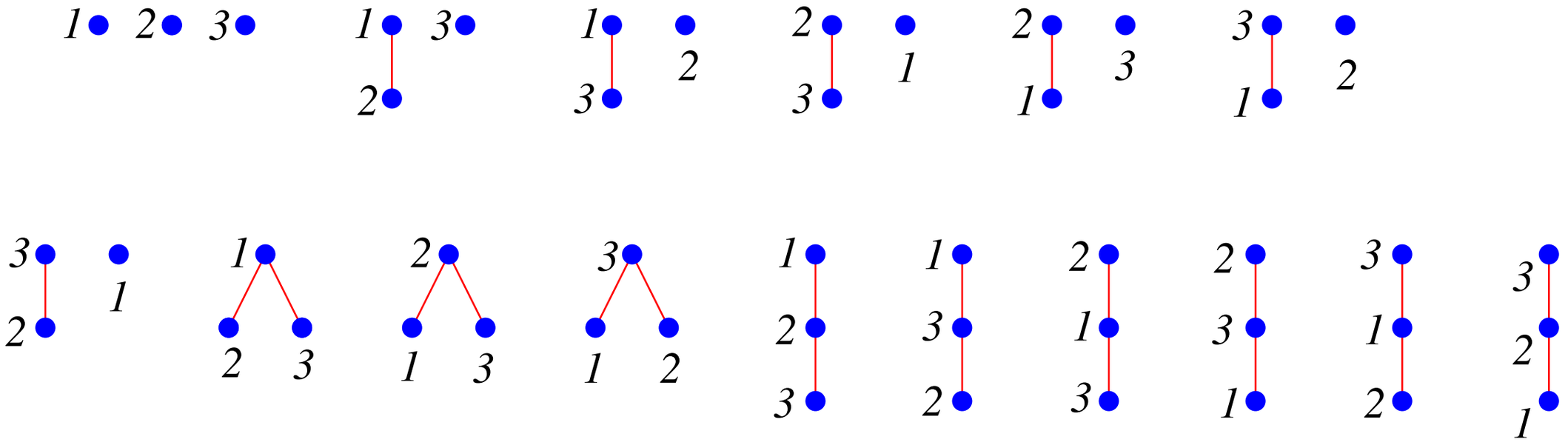
Let F be a rooted forest on the vertex set $\{1, \ldots, n\}$.



Theorem (**Sylvester-Borchardt-Cayley**). *The number of such forests is* $(n+1)^{n-1}$.

Labelled rooted forests are in bijection with parking functions

the 16 labelled rooted forests on {1,2,3}



parking functions of length 3

111 211 311 121 221 321 131 231
112 212 312 122 132
113 213 123

But what is the map??

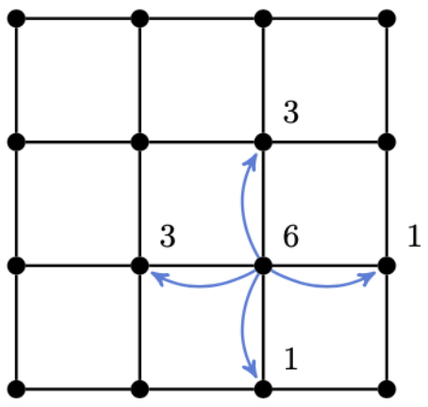
Pruffer codes…

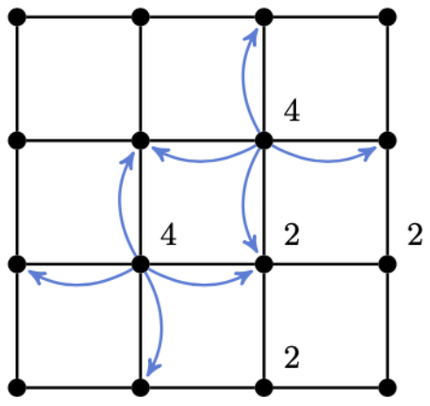
superstable chip configurations on the complete graph of order n+1

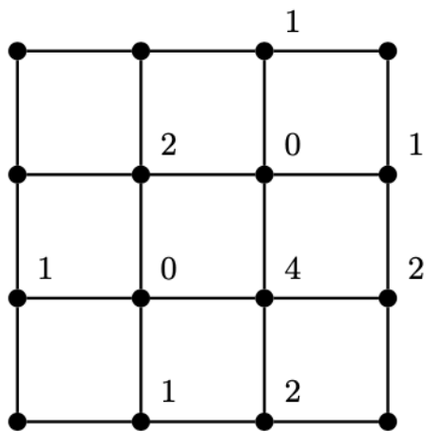
bijection

parking functions of length n

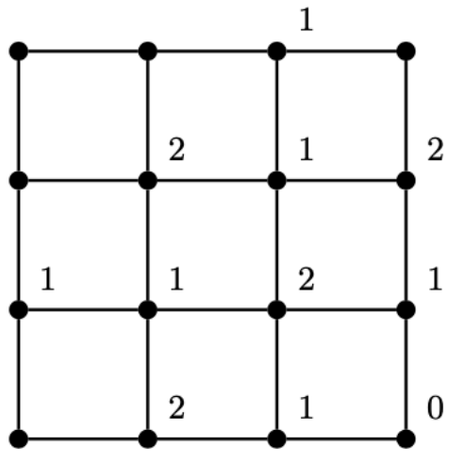
chip firing on graphs





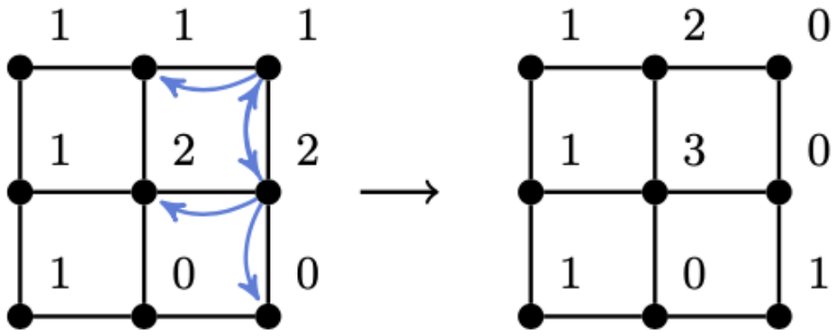


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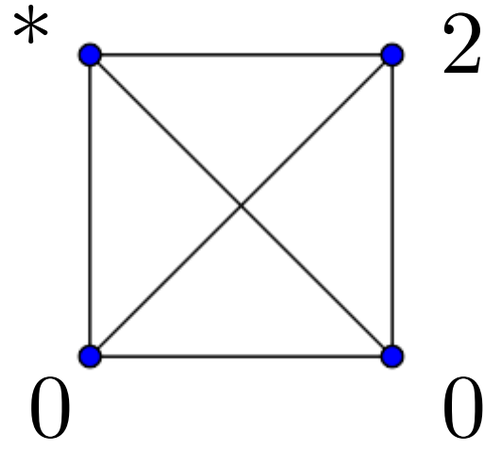


stable

a legal “cluster fire”



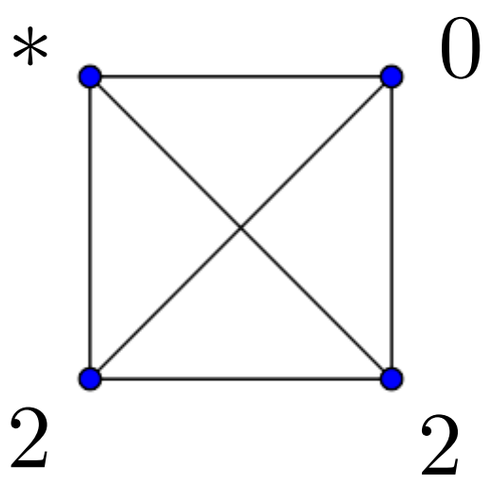
stable configuration on complete graph K4 with sink



113



stable but not super-stable



133



super-stable chip configurations on the complete graph with n+1 vertices (with sink) are in bijection with parking functions of length n

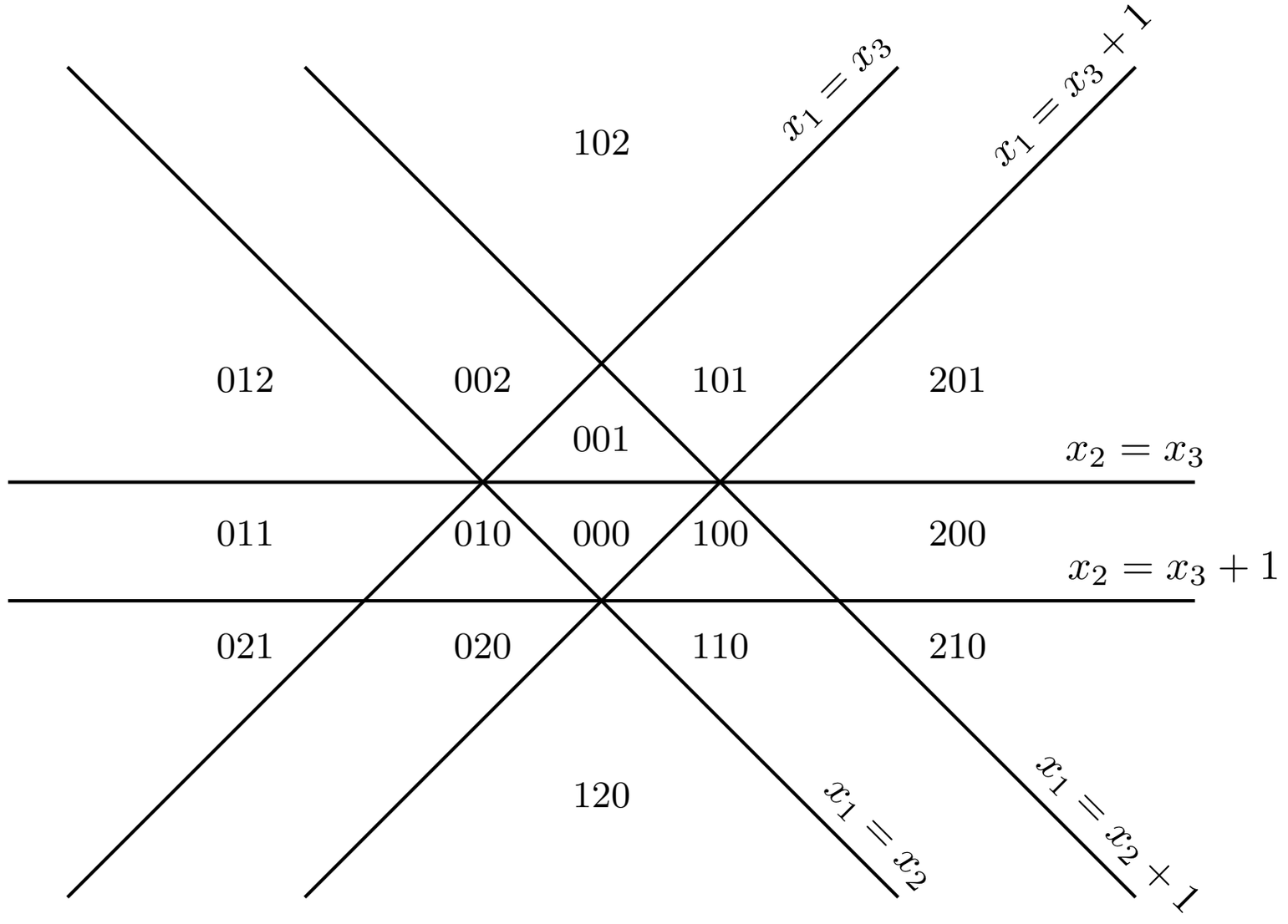
Returning to the case $G = K_n$, Pak and Stanley gave a bijective correspondence between parking functions (superstable configurations) and regions of the Shi hyperplane arrangement.

Definition 3.6.4. The *Shi arrangement* $\mathcal{A}_{\mathrm{shi}(n)}$ is the hyperplane arrangement in $\mathbb{R}^n$ consisting of the hyperplanes

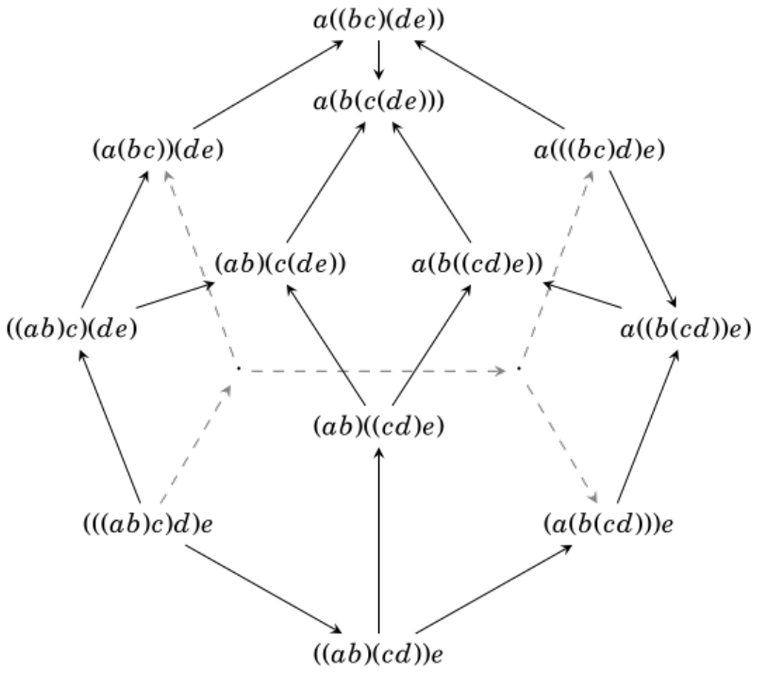
$$\{x_i = x_j, \quad x_i = x_j + 1\} \text{ for all } i < j \leq n.$$

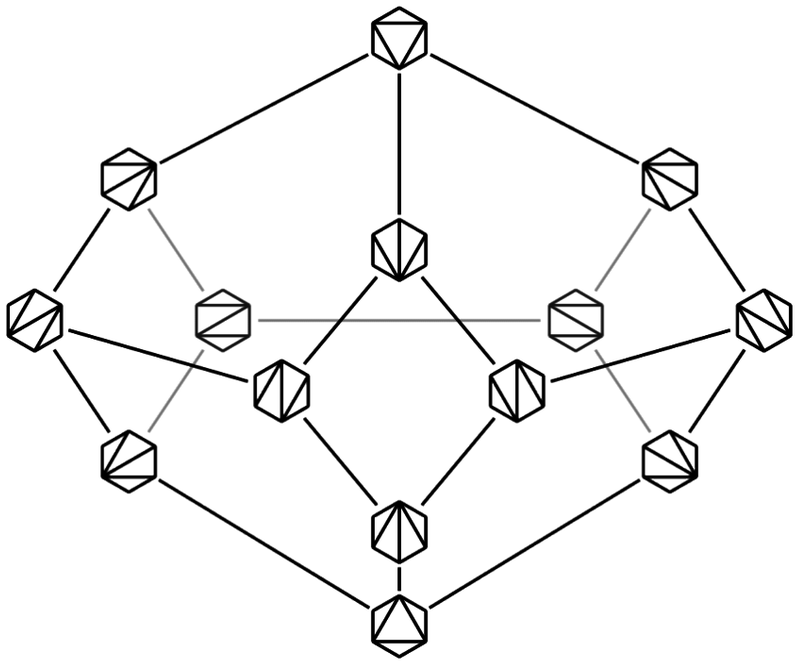
It was well known that the number of regions of the Shi arrangement is $(n+1)^{n-1}$. Pak and Stanley gave an explicit labeling of the regions of the arrangement by parking functions:

There is a bijection between certain hyperplane arrangements and parking functions



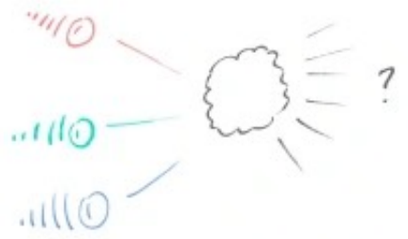
associahedron





The Amplituhedron

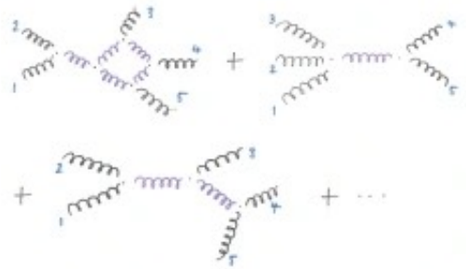
Particle scattering is a great way to learn about the universe. We can deduce properties of matter and forces by smashing together fundamental building blocks.





When particles scatter at the **LHC** in CERN we get very complicated results. Therefore we need a precise predictive theory. The best candidates are **quantum field theories** (QFTs).

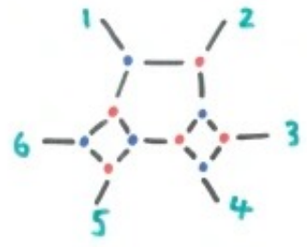
We do QFT calculations by adding up **Feynman diagrams**. Each diagram represents a possible collision with an associated probability. The simplest diagrams are known as **tree level**.

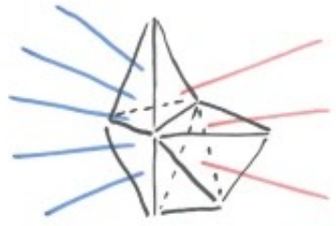




To compare theory with LHC results we must analyse the **gluons** which bind nuclei together. Typical gluon collisions involve many Feynman diagrams. Scattering ten gluons means summing up **over ten million** tree level terms!

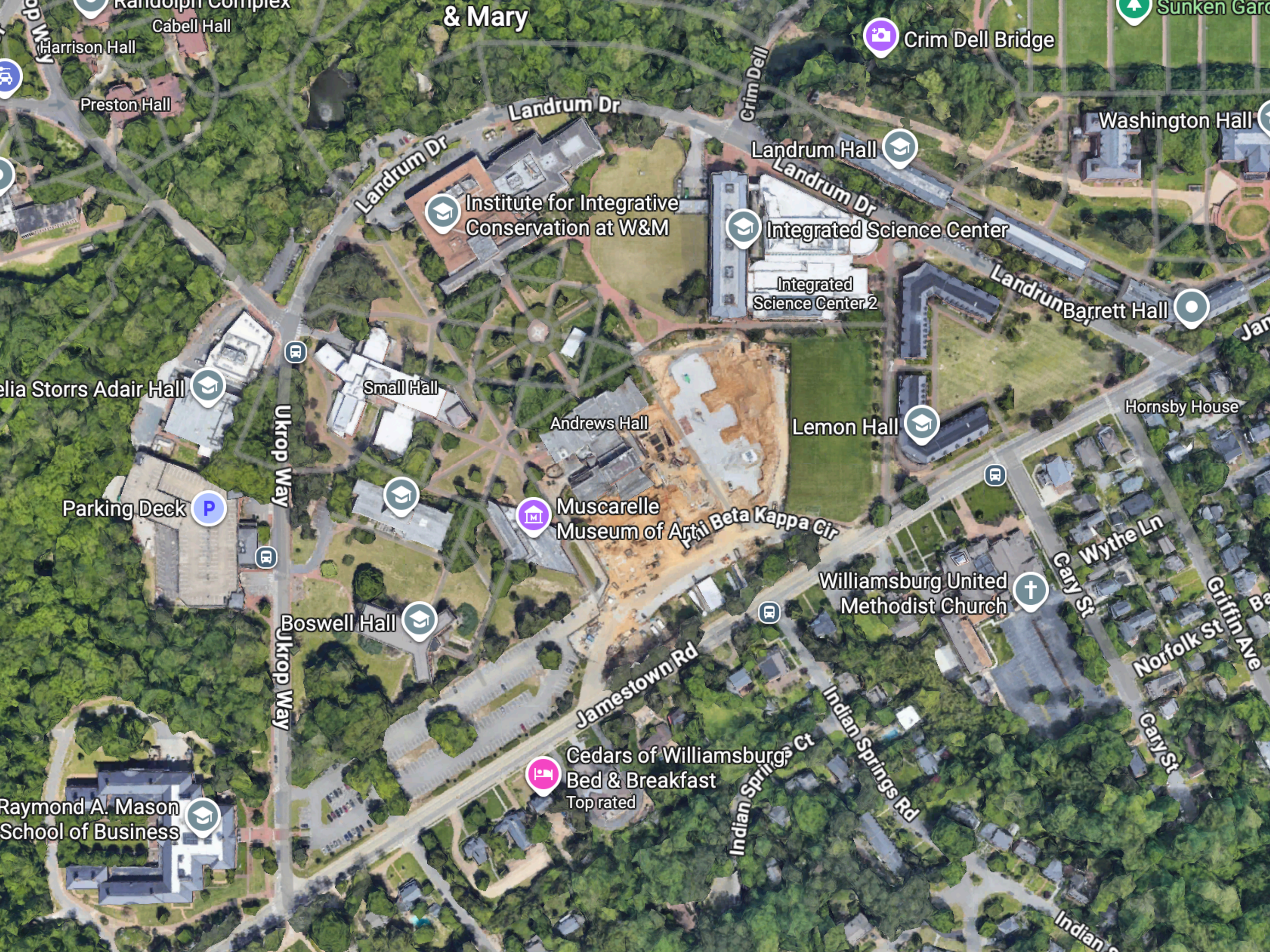
Recently **on-shell pictures** have emerged as a more efficient method. Large numbers of Feynman diagrams can be condensed into just a few on-shell pictures. That's because there's **hidden structure** not captured by Feynman's approach.



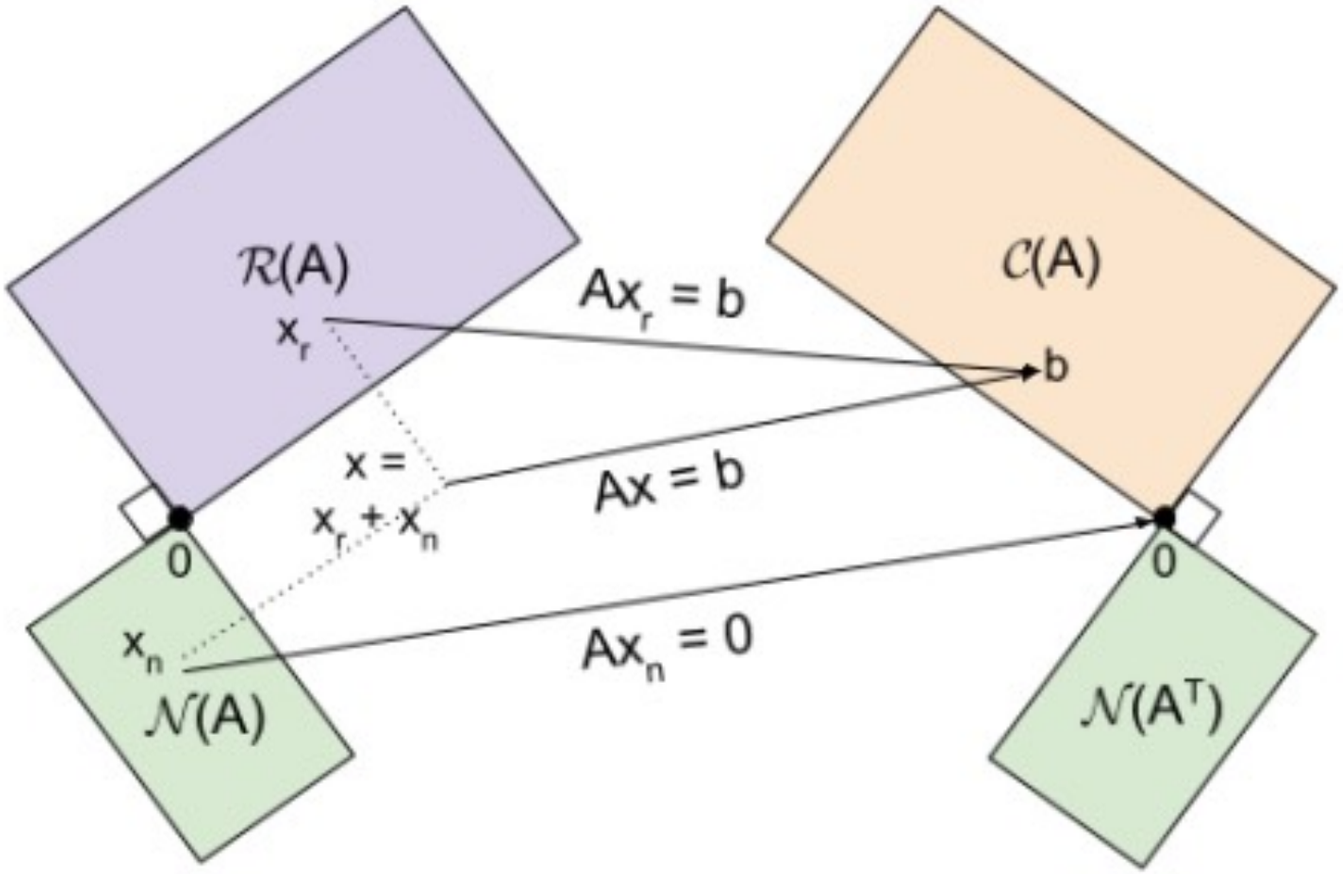


The **amplituhedron** is a single object containing all Feynman diagrams. We compute a scattering outcome merely by finding its volume. Physical laws emerge naturally from its geometry. So it might be a **gateway to new theories**.

The Amplituhedron, Trnka et al. [1312.2007]. *Scattering Amplitudes and the Positive Grassmannian*, Trnka et al. [1212.5605].


& Mary
Randolph Complex
Cabell Hall
Harrison Hall
Preston Hall
Crim Dell Bridge
Crim Dell
Landrum Dr
Landrum Dr
Landrum Dr
Landrum Hall
Washington Hall
Institute for Integrative Conservation at W&M
Integrated Science Center
Integrated Science Center 2
Barrett Hall
Small Hall
Andrews Hall
Lemon Hall
Hornsby House
Ukrop Way
Parking Deck
Muscarelle Museum of Art
Phi Beta Kappa Cir
Boswell Hall
Williamsburg United Methodist Church
Wythe Ln
Cary St
Norfolk St
Griffin Ave
Jamestown Rd
Indian Springs Ct
Indian Springs Rd
Cedars of Williamsburg Bed & Breakfast
Top rated
Raymond A. Mason School of Business

Matrix A converts n-tuples into m-tuples $\mathbb{R}^n \rightarrow \mathbb{R}^m$.
That is, linear transformation T_A is a map between rows and columns



Fundamental Subspaces

$\mathcal{C}$(A): Column space (image)
$\mathcal{R}$(A): Row space (coimage)
$\mathcal{N}$(A): Null space (kernel)
$\mathcal{N}(A^T)$: Left null space (cokernel)

Identities

$\dim(\mathcal{C}) \equiv \text{rank}(A)$
$\dim(\mathcal{N}) \equiv \text{nullity}(A)$

Theorems

$\dim(\mathcal{C}) + \dim(\mathcal{N}) = n$
$\dim(\mathcal{R}) = \dim(\mathcal{C})$

$$T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$$
$$\boldsymbol{x} \rightarrow A\boldsymbol{x}$$

